

SIMPLE LINEAR REGRESSION

Problem 1: Predicting Ice Cream Sales from Temperature

A shop records the daily temperature ($^{\circ}\text{C}$) and corresponding ice cream sales (units) for 8 days.

1. Find the linear regression equation of ice cream sales on temperature.
2. Calculate the slope and intercept.
3. Predict the ice cream sales when the temperature is 29°C .
4. Interpret the slope of the regression equation.

Given Data

Day	Temperature X ($^{\circ}\text{C}$)	Ice Cream Sales (units)
1	20	120
2	22	135
3	24	150
4	26	165
5	28	180
6	30	195
7	32	210
8	34	225

Solution

The simple linear regression model is:

$$\hat{Y} = b_0 + b_1X$$

Step 1: Construct the calculation table

X	Y	X^2	Y^2	XY
20	120	400	14400	2400
22	135	484	18225	2970
24	150	576	22500	3600
26	165	676	27225	4290
28	180	784	32400	5040
30	195	900	38025	5850
32	210	1024	44100	6720
34	225	1156	50625	7650
Σ	1380	6000	247500	38520

Therefore, $n = 8$, $\Sigma X = 216$, $\Sigma Y = 1380$, $\Sigma X^2 = 6000$, and $\Sigma XY = 38520$.

Mean temperature: $\bar{X} = \Sigma X/n = 216/8 = 27.00^{\circ}\text{C}$.

Mean response: $\bar{Y} = \Sigma Y/n = 1380/8 = 172.50$.

Step 2: Calculate the slope b_1

$$b_1 = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / [n\Sigma X^2 - (\Sigma X)^2]$$

$$b_1 = [8(38520) - (216)(1380)] / [8(6000) - (216)^2]$$

$$b_1 = 10080/1344 = 7.5000$$

Step 3: Calculate the intercept b_0

$$b_0 = \bar{Y} - b_1\bar{X}$$

$$b_0 = 172.5000 - (7.5000)(27.00) = -30.0000$$

Step 4: Regression equation

$$\hat{Y} = -30.0000 + 7.5000X$$

Step 5: Prediction for $X = 29^{\circ}\text{C}$

$$\hat{Y} = -30.0000 + (7.5000)(29) = 187.50$$

Predicted ice cream sales (units) = 187.50.

Step 6: Interpretation of the slope

The slope is 7.50. This means that for every 1°C increase in temperature, the predicted ice cream sales increase by approximately 7.50 units, on average.

Final Answer

Regression equation: $\hat{Y} = -30.0000 + 7.5000X$. At 29°C , predicted ice cream sales = 187.50 units.

Problem 2: Predicting Electricity Consumption from Temperature

An electricity distribution company records the average daily temperature and electricity consumption of a residential area.

1. Determine the simple linear regression equation.
2. Calculate the slope b_1 and intercept b_0 .
3. Predict electricity consumption when the temperature is 27°C .
4. Explain what the slope indicates about the relationship between temperature and electricity consumption.
5. Calculate the coefficient of determination R^2 and interpret it.

Given Data

Day	Temperature X ($^\circ\text{C}$)	Electricity Consumption (kWh)
1	18	210
2	20	220
3	22	235
4	24	250
5	26	270
6	28	290
7	30	315
8	32	340

Solution

The simple linear regression model is:

$$\hat{Y} = b_0 + b_1X$$

Step 1: Construct the calculation table

X	Y	X^2	Y^2	XY
18	210	324	44100	3780
20	220	400	48400	4400
22	235	484	55225	5170
24	250	576	62500	6000
26	270	676	72900	7020
28	290	784	84100	8120
30	315	900	99225	9450
32	340	1024	115600	10880
Σ	2130	5168	582050	54820

Therefore, $n = 8$, $\Sigma X = 200$, $\Sigma Y = 2130$, $\Sigma X^2 = 5168$, and $\Sigma XY = 54820$.

Mean temperature: $\bar{X} = \Sigma X/n = 200/8 = 25.00^\circ\text{C}$.

Mean response: $\bar{Y} = \Sigma Y/n = 2130/8 = 266.25$.

Step 2: Calculate the slope b_1

$$b_1 = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / [n\Sigma X^2 - (\Sigma X)^2]$$

$$b_1 = [8(54820) - (200)(2130)] / [8(5168) - (200)^2]$$

$$b_1 = 12560/1344 = 9.3452$$

Step 3: Calculate the intercept b_0

$$b_0 = \bar{Y} - b_1\bar{X}$$

$$b_0 = 266.2500 - (9.3452)(25.00) = 32.6190$$

Step 4: Regression equation

$$\hat{Y} = 32.6190 + 9.3452X$$

Step 5: Prediction for $X = 27^\circ\text{C}$

$$\hat{Y} = 32.6190 + (9.3452)(27) = 284.94$$

Predicted electricity consumption (kwh) = 284.94.

Step 6: Interpretation of the slope

The slope is 9.35. This means that for every 1°C increase in temperature, the predicted electricity consumption increases by approximately 9.35 kWh, on average.

Step 7: Coefficient of determination R^2

$$R^2 = 0.9822 = 98.22\%.$$

Interpretation: Approximately 98.22% of the variation in electricity consumption is explained by temperature through this linear regression model. The remaining 1.78% is associated with other factors and random variation.

Final Answer

Regression equation: $\hat{Y} = 32.6190 + 9.3452X$. At 27°C , predicted electricity consumption = 284.94 kWh, with $R^2 = 0.9822$.

Multiple linear regression problems

Problem 1: House Price Prediction

Given data

Let:

X_1 = Area in 100 sq.ft.

X_2 = Number of bedrooms

Y = House price in ₹ lakh

House	X_1	X_2	Y
1	8	2	48
2	10	3	58
3	12	3	68
4	14	4	78
5	16	4	88
6	18	5	98

We assume the multiple linear regression model:

$$Y = a + b_1X_1 + b_2X_2$$

where:

a = intercept

b_1 = least-squares estimate for X_1

b_2 = least-squares estimate for X_2

Step 1: Calculate the required sums

We need:

$$\begin{aligned} &\sum Y, \sum X_1, \sum X_2 \\ &\sum X_1^2, \sum X_2^2, \sum X_1 X_2 \\ &\sum X_1 Y, \sum X_2 Y \end{aligned}$$

The calculation table is:

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
8	2	48	64	4	16	384	96
10	3	58	100	9	30	580	174
12	3	68	144	9	36	816	204
14	4	78	196	16	56	1092	312
16	4	88	256	16	64	1408	352
18	5	98	324	25	90	1764	490
Σ							

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
78	21	438	1084	79	292	6044	1628

Therefore:

$$\begin{aligned}
 n &= 6 \\
 \sum X_1 &= 78 \\
 \sum X_2 &= 21 \\
 \sum Y &= 438 \\
 \sum X_1^2 &= 1084 \\
 \sum X_2^2 &= 79 \\
 \sum X_1 X_2 &= 292 \\
 \sum X_1 Y &= 6044 \\
 \sum X_2 Y &= 1628
 \end{aligned}$$

Step 2: Form the least-squares normal equations

For

$$Y = a + b_1X_1 + b_2X_2$$

the normal equations are:

$$\begin{aligned}
 \sum Y &= na + b_1 \sum X_1 + b_2 \sum X_2 \\
 \sum X_1 Y &= a \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2 \\
 \sum X_2 Y &= a \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2
 \end{aligned}$$

Substituting the values:

Equation 1

$$438 = 6a + 78b_1 + 21b_2$$

Equation 2

$$6044 = 78a + 1084b_1 + 292b_2$$

Equation 3

$$1628 = 21a + 292b_1 + 79b_2$$

Thus, we have three simultaneous equations:

$$6a + 78b_1 + 21b_2 = 438 \quad (1)$$

$$78a + 1084b_1 + 292b_2 = 6044 \quad (2)$$

$$21a + 292b_1 + 79b_2 = 1628 \quad (3)$$

Step 3: Solve the normal equations

From Equation (1):

$$6a + 78b_1 + 21b_2 = 438$$

Divide by 6:

$$a + 13b_1 + 3.5b_2 = 73$$

Therefore,

$$a = 73 - 13b_1 - 3.5b_2$$

Substitute this into Equations (2) and (3).

After simplification:

$$70b_1 + 19b_2 = 350 \quad (4)$$

$$19b_1 + 5.5b_2 = 95 \quad (5)$$

Multiply Equation (5) by 2:

$$38b_1 + 11b_2 = 190 \quad (6)$$

Solving Equations (4) and (6):

$$70b_1 + 19b_2 = 350$$

$$38b_1 + 11b_2 = 190$$

The solution is:

$$b_1 = 5$$

and

$$b_2 = 0$$

Now:

$$a = 73 - 13(5) - 3.5(0)$$

$$a = 73 - 65$$

$$a = 8$$

Least-squares regression equation

Therefore,

$$\hat{Y} = 8 + 5X_1 + 0X_2$$

or simply,

$$\hat{Y} = 8 + 5X_1$$

Interpretation

$b_1 = 5$: For every additional 100 sq.ft. of area, predicted house price increases by ₹5 lakh, holding the number of bedrooms constant.

$b_2 = 0$: For this particular dataset, bedrooms do not contribute an additional amount after accounting for area.

Step 4: Prediction

For:

$$X_1 = 15, X_2 = 4$$

$$\hat{Y} = 8 + 5(15) + 0(4)$$

$$= 8 + 75$$

$$\hat{Y} = 83 \text{ lakh}$$

Problem 2: Student Marks Prediction

Given data

Let:

X_1 = Study hours/week

X_2 = Number of practice tests

Y = Examination marks

Student	X_1	X_2	Y
1	2	1	52
2	3	1	57
3	4	2	66
4	5	2	71
5	6	3	80
6	7	3	85

We use:

$$Y = a + b_1X_1 + b_2X_2$$

Step 1: Calculate the required values

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
2	1	52	4	1	2	104	52
3	1	57	9	1	3	171	57
4	2	66	16	4	8	264	132
5	2	71	25	4	10	355	142
6	3	80	36	9	18	480	240
7	3	85	49	9	21	595	255
Σ							
27	12	411	139	28	62	1969	878

Thus:

$$\begin{aligned}n &= 6 \\ \sum X_1 &= 27, \quad \sum X_2 = 12, \quad \sum Y = 411 \\ \sum X_1^2 &= 139, \quad \sum X_2^2 = 28 \\ \sum X_1 X_2 &= 62 \\ \sum X_1 Y &= 1969, \quad \sum X_2 Y = 878\end{aligned}$$

Step 2: Construct the normal equations

Using:

$$\sum Y = na + b_1 \sum X_1 + b_2 \sum X_2$$

we get:

$$411 = 6a + 27b_1 + 12b_2 \quad (1)$$

The second normal equation is:

$$1969 = 27a + 139b_1 + 62b_2 \quad (2)$$

The third normal equation is:

$$878 = 12a + 62b_1 + 28b_2 \quad (3)$$

Therefore:

$$\begin{aligned} 6a + 27b_1 + 12b_2 &= 411 \\ 27a + 139b_1 + 62b_2 &= 1969 \\ 12a + 62b_1 + 28b_2 &= 878 \end{aligned}$$

Step 3: Solve the equations

From Equation (1):

$$6a + 27b_1 + 12b_2 = 411$$

Therefore:

$$\begin{aligned} a &= \frac{411 - 27b_1 - 12b_2}{6} \\ a &= 68.5 - 4.5b_1 - 2b_2 \end{aligned}$$

Substituting into Equations (2) and (3), we obtain:

$$17.5b_1 + 8b_2 = 119.5 \quad (4)$$

and

$$8b_1 + 4b_2 = 55 \quad (5)$$

From Equation (5):

$$\begin{aligned} 8b_1 + 4b_2 &= 55 \\ 4b_2 &= 55 - 8b_1 \\ b_2 &= 13.75 - 2b_1 \end{aligned}$$

Substitute into Equation (4):

$$\begin{aligned} 17.5b_1 + 8(13.75 - 2b_1) &= 119.5 \\ 17.5b_1 + 110 - 16b_1 &= 119.5 \\ 1.5b_1 &= 9.5 \end{aligned}$$

This gives:

$$b_1 = 6.3333$$

However, this indicates an inconsistency with the simplified equations above due to rounding/arithmetic. Using the original normal equations directly gives the exact least-squares solution:

$$\boxed{a = 38}$$

$$\boxed{b_1 = 5}$$

$$\boxed{b_2 = 4}$$

Hence the correct regression equation is:

$$\hat{Y} = 38 + 5X_1 + 4X_2$$

Step 4: Verify the equation

For Student 1:

$$\begin{aligned}\hat{Y} &= 38 + 5(2) + 4(1) \\ &= 38 + 10 + 4 = 52\end{aligned}$$

Actual:

$$Y = 52$$

For Student 3:

$$\begin{aligned}\hat{Y} &= 38 + 5(4) + 4(2) \\ &= 38 + 20 + 8 = 66\end{aligned}$$

Actual:

$$Y = 66$$

Similarly, the equation reproduces all six observations exactly.

Step 5: Prediction

Given:

$$X_1 = 5.5$$

and

$$X_2 = 3$$

Then:

$$\begin{aligned}\hat{Y} &= 38 + 5(5.5) + 4(3) \\ &= 38 + 27.5 + 12\end{aligned}$$

$$\hat{Y} = 77.5$$

So the predicted examination mark is:

$$77.5\%$$